THE NUMBER $3^{3^{18}}+2$ IS DIVISIBLE BY 31^2: A COUNTEREXAMPLE TO HILLIARD'S SQUAREFREE CONJECTURE IN OEIS A083020

MATH0

Abstract. The OEIS entry A083020 (Cino Hilliard, 2003) lists the prime factors of the numbers $N_n = 3^{3^n}+2$, $n \geq 0$, and conjectures that every N_n is squarefree. We show that 31^2 divides N_{18}, a number with 184,846,550 decimal digits, so the conjecture is false; the proof reduces the exponent 3^{18} modulo $\varphi(31^2) = 930$ and then needs only the congruence $3^{159} \equiv -2 \pmod{961}$. We also show, without formalizing it, that $31^2 \mid N_n$ exactly when $n \equiv 18 \pmod{60}$ and $29^2 \mid N_n$ exactly when $n \equiv 73 \pmod{84}$, so infinitely many N_n are not squarefree. To our knowledge the counterexample is new: the OEIS entry still states the conjecture and we found no recorded resolution, although the argument is elementary. The refutation is verified in Lean 4 with Mathlib against the frozen statement that math0 formalized from the OEIS entry; its definitions are checked by the kernel against the entry's data, and it was reviewed for fidelity before the proof attempt.

1. Introduction

For $n \geq 0$ let

$$N_n = 3^{3^n}+2.$$

The OEIS entry A083020 [6], submitted by Cino Hilliard on May 31, 2003, is the irregular triangle whose row n lists the prime factors of N_n in increasing order. Its example section reads $N_0 = 5$, $N_1 = 29$, $N_2 = 5 \cdot 31 \cdot 127$, $N_3 = 11 \cdot 693236134999$; the listed terms continue with the six prime factors of $N_4 = 3^{81}+2$, the 116-digit number $N_5 = 3^{243}+2$ (so the entry asserts that N_5 is prime), and the two smallest prime factors $5, 31$ of the 348-digit number N_6. The comments contain the conjecture that all of these numbers are squarefree (Conjecture 2 below).

We show that the conjecture fails at $n = 18$.

Theorem 1. $31^2 = 961$ *divides* $N_{18} = 3^{3^{18}}+2$. *In particular* N_{18} *is not squarefree.*

The number N_{18} has $\lfloor 3^{18} \log_{10} 3 \rfloor + 1 = 184{,}846{,}550$ decimal digits, so it is far out of reach of factorization; but divisibility by the fixed number 961 only depends on the residue of $3^{3^{18}}$ modulo 961, which Euler's theorem reduces to the residue of 3^{159} (Section 3). The same reduction describes all n (Proposition 6): $961 \mid N_n$ if and only if $n \equiv 18 \pmod{60}$, and $29^2 = 841$ divides N_n if and only if $n \equiv 73 \pmod{84}$. In particular infinitely many N_n are not squarefree.

Status and prior work. The entry A083020 (revision #20 of January 4, 2026, which is still the current revision in October 2026) still states the conjecture; its EXTENSIONS section records only a correction by T. D. Noe (November 15, 2006) and a clarified name with the terms $a(14), a(15)$ added by Sean A. Irvine (January 4, 2026). The math0 literature referees found no prior counterexample or proof. They checked the OEIS entries on related numbers, namely A055777 (3^{3^n}) [4], A129290 ($3^{3^n}+1$) [8], A129291 [9], A113913 [7], A140331 [10] and A061678 [5], none of which discusses the squarefreeness of N_n; Stephan's collection *Prove or disprove. 100 conjectures from the OEIS* [11], which does not contain A083020; and web searches of arXiv, MathOverflow, Mathematics Stack Exchange and the SeqFan mailing list. The conjecture is not

Date: October 21, 2026.

among the OEIS statements of the formal-conjectures repository [1] at the commit pinned by math0 (`c77fdd4`). We therefore classify Theorem 1 as a new result, with the caveat that the argument is elementary and the counterexample may have been noticed without being recorded. We do not claim that $n = 18$ is the least n for which N_n is not squarefree (Remark 7).

2. Statement

The OEIS conjecture. The COMMENTS section of A083020 contains the following sentence, verbatim:

> `Conjecture: These numbers have no repeating factors or 3^(3^n) + 2 is squarefree.`

The comment carries no separate attribution; it is part of the original entry by Cino Hilliard, May 31, 2003. The entry has offset 0, so "these numbers" are N_n for all $n \geq 0$, and the two phrasings joined by "or" say the same thing: N_n has no repeated prime factor.

Conjecture 2 (Hilliard)**.** *For every integer $n \geq 0$, the number $N_n = 3^{3^n} + 2$ is squarefree.*

By Theorem 1, Conjecture 2 is false.

Lean statement. The statement is the definition `Math0.Statements.A083020.statement` in the math0 statement file `lean/Math0/Statements/OEIS/A083020.lean`, which was reviewed for fidelity and frozen (SHA-256 `d520c666...ae473`) before any proof attempt. The definitions that encode the triangle, and the statement, read verbatim (docstrings omitted; namespace `Math0.Statements.A083020`):

```
def N (n : ℕ) : ℕ := 3 ^ (3 ^ n) + 2

def row (n : ℕ) : List ℕ := Nat.primeFactorsList (N n)

def a (i : ℕ) : ℕ := ((List.range (i + 1)).flatMap row).getD i 0

def statement : Prop :=
  ∀ n : ℕ, Squarefree (3 ^ (3 ^ n) + 2)
```

In Mathlib, `Squarefree r` is defined as $\forall x,\ x \cdot x \mid r \to$ `IsUnit` x, and the only unit of $\mathbb{N}$ is 1 (`Nat.isUnit_iff`); so for a natural number r it says that no square x^2 with $x \geq 2$ divides r, which is the usual notion. Thus `statement` is Conjecture 2, quantified over all $n \in \mathbb{N}$ as the offset 0 requires, with no extra hypotheses; it involves no subtraction or division in $\mathbb{N}$, and $N_n \geq 5$ rules out degenerate cases. The file also proves

```
theorem statement_iff : statement ↔ ∀ n, (row n).Nodup
```

from Mathlib's `Nat.squarefree_iff_nodup_primeFactorsList`; this is the "no repeating factors" phrasing: no prime occurs twice in row n. We prove ¬ `statement`.

Encoding and check against the OEIS data. Row n of the triangle is `row n`, Mathlib's list of the prime factors of N_n in increasing order with multiplicity. The listed rows contain no repeated primes, so listing distinct primes instead would give the same terms; the conjecture is stated directly about N_n in any case. The sequence itself is `a`, the triangle read by rows with offset 0: `a i` is the entry at position i of the concatenation of rows $0, \ldots, i$, which has at least $i + 1$ entries because every row is nonempty.

The function `a` is computable, but evaluating it means factoring N_n by trial division, so the statement file checks the data through certificates instead. The lemma `primeFactorsList_eq_`

`of` says that a sorted list of primes with product m is `Nat.primeFactorsList m` (uniqueness of factorization), and `row_data` applies it to

$$5, \quad 29, \quad 5\cdot 31\cdot 127, \quad 11\cdot 693236134999, \quad 5\cdot 15679\cdot 2372959\cdot 5945531\cdot 3312453269\cdot 121032673159$$

for $n = 0, \dots, 4$. The products are checked by `norm_num`, sortedness by `decide`, primes below 10^5 by `norm_num`, and the larger primes by Lucas certificates (Table 1). Mathlib's `lucas_primality` says that p is prime if some w satisfies $w^{p-1} \equiv 1 \pmod p$ and $w^{(p-1)/q} \not\equiv 1 \pmod p$ for every prime $q \mid p-1$. The Boolean function `lucasCheck` tests these conditions for a given w and a list of primes with product $p-1$, computing powers with a square-and-multiply function `powMod` that is proved equal to $w^e \bmod p$ for $e < 2^{64}$; the theorem `prime_of_lucasCheck` turns a successful check into a proof of primality, and each check is evaluated by `decide +kernel`. The primes in the lists for $p-1$ are proved prime by `norm_num` or by certificates of their own. The theorem `a_data` then states that $a(0), \dots, a(12)$ are the 13 listed terms $5, 29, 5, 31, 127, 11, 693236134999, 5, 15679, 2372959, 5945531, 3312453269, 121032673159$.

p	w	factorization of $p-1$
391151	13	$2 \cdot 5^2 \cdot 7823$
693236134999	7	$2 \cdot 3^2 \cdot 11 \cdot 8951 \cdot 391151$
2372959	3	$2 \cdot 3^2 \cdot 7 \cdot 37 \cdot 509$
5945531	2	$2 \cdot 5 \cdot 37 \cdot 16069$
22381441	7	$2^7 \cdot 3 \cdot 5 \cdot 11657$
3312453269	2	$2^2 \cdot 37 \cdot 22381441$
4317661	6	$2^2 \cdot 3^2 \cdot 5 \cdot 17^2 \cdot 83$
20172112193	3	$2^6 \cdot 73 \cdot 4317661$
121032673159	3	$2 \cdot 3 \cdot 20172112193$

TABLE 1. The Lucas certificates in the statement file: for each p, the witness w and the prime factors of $p-1$.

The remaining listed terms are $a(13)$, a 116-digit number, and $a(14) = 5$, $a(15) = 31$. The theorem `N5_data` checks that $a(13)$ as listed equals $N_5 = 3^{243} + 2$, and `row6_data` checks that row 6 begins with $5, 31$, by computing the two smallest prime factors of N_6. The primality of N_5 is not proved in Lean (SymPy's primality test reports it prime), so $a(13), a(14), a(15)$ sit at the listed positions exactly when N_5 is prime, and these positions are not checked as indices. All of these checks live in the statement file; the refutation below does not depend on them.

3. Proof

The proof follows the declarations of the Lean file: the private lemmas `pow_930`, `split_exp` and `sq_dvd`, and the theorem `result`. Congruences are in $\mathbb{N}$ (`Nat.ModEq`).

Lemma 3 (`pow_930`)**.** $3^{930} \equiv 1 \pmod{961}$.

Proof. Since 31 is prime, $\varphi(31^2) = 31 \cdot 30 = 930$, and $\gcd(3, 961) = 1$, so this is Euler's theorem. The Lean proof does not use Euler's theorem: it unfolds the congruence to $3^{930} \bmod 961 = 1 \bmod 961$ and checks this equation by `decide +kernel`, which makes the kernel evaluate both sides; 3^{930} has 1475 binary digits. □

Lemma 4 (`split_exp`)**.** $3^{18} = 930 \cdot 416581 + 159$*, and hence* $3^{3^{18}} = \left(3^{930}\right)^{416581} \cdot 3^{159}$.

Proof. $3^{18} = 387420489$ and $930 \cdot 416581 = 387420330$. The second claim follows from the first by the laws $(x^b)^c = x^{bc}$ and $x^b x^c = x^{b+c}$. In Lean, the right-hand side is rewritten with these

two laws (`pow_mul`, `pow_add`) into $3^{930\cdot 416581+159}$, and `norm_num` checks that the two exponents agree; the number $3^{3^{18}}$ is never evaluated. □

Lemma 5 (`sq_dvd`)**.** $31 \cdot 31$ *divides* $3^{3^{18}} + 2$.

Proof. Raising Lemma 3 to the power 416581, multiplying by 3^{159} on the right and adding 2 gives

$$\left(3^{930}\right)^{416581} \cdot 3^{159} + 2 \equiv 1^{416581} \cdot 3^{159} + 2 \pmod{961}$$

(`Nat.ModEq.pow`, `Nat.ModEq.mul_right`, `Nat.ModEq.add_right`). Next, $3^{159} \equiv 959 \equiv -2 \pmod{961}$, so $1^{416581} \cdot 3^{159} + 2 \equiv 961 \equiv 0 \pmod{961}$. To check the residue by hand: the successive squares $3^{2^k} \bmod 961$ for $k = 0, \ldots, 7$ are

$$3,\ 9,\ 81,\ 795,\ 648,\ 908,\ 887,\ 671,$$

and $159 = 2^7+2^4+2^3+2^2+2+1$, so $3^{159} \equiv 3\cdot 9\cdot 81\cdot 795\cdot 648\cdot 671$; the partial products reduce to $27,\ 265,\ 216,\ 623,\ 959$ modulo 961. In Lean, the congruence $1^{416581} \cdot 3^{159} + 2 \equiv 0 \pmod{961}$ is checked by `decide +kernel`. By transitivity and Lemma 4, $3^{3^{18}} + 2 \equiv 0 \pmod{961}$, that is, 961 divides $3^{3^{18}} + 2$ (`Nat.modEq_zero_iff_dvd`). In Lean the goal $31 \cdot 31 \mid \ldots$ is first rewritten with Lemma 4, and $31 \cdot 31$ is identified with 961 by evaluating the product of the two literals. □

As a consistency check, which the proof does not use: modulo 31 we have $3^{30} \equiv 1$ and $3^{18} \equiv 9 \pmod{30}$, so $3^{3^{18}} \equiv 3^9 \equiv 29 \equiv -2 \pmod{31}$, as Lemma 5 requires.

Proof of Theorem 1 and of ¬ *`statement (result)`.* Lemma 5 is the first claim. Suppose that `statement` holds. Then N_{18} is squarefree, so by the definition of `Squarefree`, applied to $x = 31$ and Lemma 5, 31 is a unit of $\mathbb{N}$. By `Nat.isUnit_iff` this means $31 = 1$, which is false (`omega`). □

All n, and small primes. The following is not part of the formal proof.

Proposition 6. *For $n \geq 0$: $961 \mid N_n$ if and only if $n \equiv 18 \pmod{60}$, and $841 \mid N_n$ if and only if $n \equiv 73 \pmod{84}$. In particular infinitely many N_n are not squarefree.*

Proof. Let $x_n = 3^{3^n} \bmod 961$, so that $x_{n+1} \equiv x_n^3 \pmod{961}$, and $961 \mid N_n$ if and only if $x_n = 959$. By Lemma 3, x_n depends only on $3^n \bmod 930$. Now $930 = 3\cdot 310$ with $\gcd(3, 310) = 1$. For $n \geq 1$ we have $3^n \equiv 0 \pmod 3$, and $3^n \bmod 310$ is periodic in n with period the order of 3 modulo $310 = 2 \cdot 5 \cdot 31$, which is $\operatorname{lcm}(1, 4, 30) = 60$. By the Chinese remainder theorem, $3^n \bmod 930$, and hence x_n, depends only on $n \bmod 60$ for $n \geq 1$. Computing $x_1, \ldots, x_{60}$ by repeated cubing modulo 961 gives $x_n = 959$ only for $n = 18$, and $x_0 = 3$. This proves the first claim. For $841 = 29^2$ we have $\varphi(841) = 812 = 2^2 \cdot 7 \cdot 29$, which is prime to 3, so $3^n \bmod 812$ is purely periodic in n with period the order of 3 modulo 812, which is $\operatorname{lcm}(2, 6, 28) = 84$. Hence $y_n = 3^{3^n} \bmod 841$ is periodic with period dividing 84, and computing $y_0, \ldots, y_{83}$ gives $y_n = 839$ only for $n = 73$. □

Remark 7 (Smaller n)**.** For $n \leq 4$ the factorizations in `row_data` show that N_n is squarefree, and N_5 is squarefree if it is prime, as the entry asserts. For $6 \leq n \leq 17$ we do not know whether N_n is squarefree: these numbers have between 348 and about 6.2×10^7 decimal digits, and none of them is completely factored (for N_6 only the factors 5 and 31 and a composite cofactor are known). A computer search (not part of the formal proof) that computes $3^{3^n} \bmod p^2$ by repeated cubing, for every prime $p < 10^8$ and every $n \leq 18$, finds $p^2 \mid N_n$ only for $(n, p) = (18, 31)$. So $n = 18$ is the least n for which N_n is divisible by the square of a prime below 10^8, but we do not know whether it is the least n for which N_n is not squarefree.

4. Machine verification

Files and declarations. The proof is in the file `lean/Math0/Frontier/OEIS/A083020.lean` of the math0 repository, `https://github.com/Foundation-0/math0`. It imports the frozen statement file `lean/Math0/Statements/OEIS/A083020.lean` (Section 2) and the audit commands in `lean/Math0/Audit.lean`. The main declaration is `Math0.Frontier.A083020.result`, of type `¬ Math0.Statements.A083020.statement`; the auxiliary lemmas `pow_930`, `split_exp` and `sq_dvd` are private. The proof file reads, verbatim except for the license header, the imports, the docstrings and blank lines:

```
namespace Math0.Frontier.A083020
open Math0.Statements.A083020
private theorem pow_930 : 3 ^ 930 ≡ 1 [MOD 961] := by
  unfold Nat.ModEq
  decide +kernel
private theorem split_exp : (3 : ℕ) ^ (3 ^ 18) = (3 ^ 930) ^ 416581 * 3 ^ 159 := by
  rw [← pow_mul, ← pow_add]
  norm_num
private theorem sq_dvd : 31 * 31 ∣ 3 ^ (3 ^ 18) + 2 := by
  have h : (3 ^ 930) ^ 416581 * 3 ^ 159 + 2 ≡ 1 ^ 416581 * 3 ^ 159 + 2 [MOD 961] :=
    ((pow_930.pow 416581).mul_right _).add_right _
  have h0 : 1 ^ 416581 * 3 ^ 159 + 2 ≡ 0 [MOD 961] := by
    unfold Nat.ModEq
    decide +kernel
  rw [split_exp]
  exact Nat.modEq_zero_iff_dvd.1 (h.trans h0)
theorem result : ¬ statement := by
  intro h
  have hu : IsUnit (31 : ℕ) := h 18 31 sq_dvd
  rw [Nat.isUnit_iff] at hu
  omega
end Math0.Frontier.A083020
#math0_settles Math0.Frontier.A083020.result Math0.Statements.A083020.statement
```

The file uses no `sorry`, `native_decide`, `set_option` or macros.

How the kernel checks the computations. The tactic `decide +kernel` proves a decidable proposition by asking the kernel to reduce its decision procedure to `true`. Lean's kernel evaluates arithmetic on natural number literals (addition, multiplication, powers, division and remainder, among others) with arbitrary-precision integer arithmetic, so the checks $3^{930} \bmod 961 = 1$ and $(1^{416581} \cdot 3^{159} + 2) \bmod 961 = 0$ are immediate; unlike `native_decide`, this adds no axiom. The only large exponent is handled symbolically: the proof manipulates `Nat.ModEq` terms whose sides are never evaluated, and Lemma 4 only compares the exponents 3^{18} and $930 \cdot 416581 + 159$. The data checks in the statement file (Section 2) use `decide +kernel` in the same way, on the Lucas certificates.

Audit. The command `#math0_settles`, defined in `lean/Math0/Audit.lean` for statements that math0 formalizes itself, fails unless (1) `statement` is a `Prop`-valued definition without universe parameters; (2) both `statement` and `result` use no axioms besides `propext`, `Classical.choice` and `Quot.sound`, which rejects `sorryAx`, the axiom `Lean.ofReduceBool` introduced

by `native_decide`, and custom axioms; and (3) the type of `result` is syntactically the statement or its negation, up to metadata annotations, with no unfolding. Running `lake env lean Math0/Frontier/OEIS/A083020.lean` in the directory `lean` prints

```
math0 audit: 'Math0.Statements.A083020.statement' depends on axioms [propext]
math0 audit: 'Math0.Frontier.A083020.result' depends on axioms [Quot.sound, propext]
math0 statement check: 'Math0.Frontier.A083020.result' refutes 'Math0.Statements.A083020.statement'
```

so `result` proves the negation of the statement and depends only on `propext` and `Quot.sound`, a subset of the three permitted axioms.

Toolchain and reproduction. We use Lean 4 [3] `v4.33.1` and Mathlib [2] `v4.33.1`, pinned in `lean/`; the repository also pins formal-conjectures [1] at commit `c77fdd4`, which this result does not use. To reproduce, run `scripts/verify.sh` from the repository root: it fetches the Mathlib build cache, builds all math0 results with `lake build` (re-running every audit command), checks that the pinned `formal_conjectures` and `mathlib` checkouts are unmodified, and reports the number of passed statement checks. To check this result alone, run `lake build Math0.Statements.OEIS.A083020` and then `lake env lean Math0/Frontier/OEIS/A083020.lean` in the directory `lean`; the kernel computations take a few seconds, and most of the running time is spent loading Mathlib.

References

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[2] The mathlib Community, *The Lean mathematical library*, in: Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020), ACM, 2020, 367–381.

[3] L. de Moura and S. Ullrich, *The Lean 4 theorem prover and programming language*, in: Automated Deduction – CADE 28, Lecture Notes in Comput. Sci. **12699**, Springer, 2021, 625–635.

[4] The OEIS Foundation Inc., `https://oeis.org/A055777` (3^{3^n}), entry by H. Bottomley, Jul. 12, 2000.

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[6] The OEIS Foundation Inc., `https://oeis.org/A083020` (prime factors of $3^{3^n}+2$), entry by C. Hilliard, May 31, 2003; revision #20, Jan. 4, 2026.

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[8] The OEIS Foundation Inc., `https://oeis.org/A129290` (base-3 Fermat numbers $3^{3^n}+1$), entry by A. Adamchuk, Apr. 8, 2007.

[9] The OEIS Foundation Inc., `https://oeis.org/A129291` ($1-3^{3^n}+9^{3^n}$), entry by A. Adamchuk, Apr. 8, 2007.

[10] The OEIS Foundation Inc., `https://oeis.org/A140331` (smallest k such that $3^{3^n}-k$ is prime), entry by J. Vos Post, May 26, 2008.

[11] R. Stephan, *Prove or disprove. 100 conjectures from the OEIS*, arXiv:math/0409509, 2004.